VRIFA

VARTM Resin Infusion Flow-Front Assessment

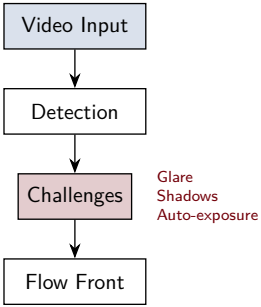
Automated Flow-Front Detection for Composite Manufacturing

Computer Vision for VARTM Process Monitoring

The Challenge: VARTM Process Monitoring

Vacuum-Assisted Resin Transfer Molding (VARTM)

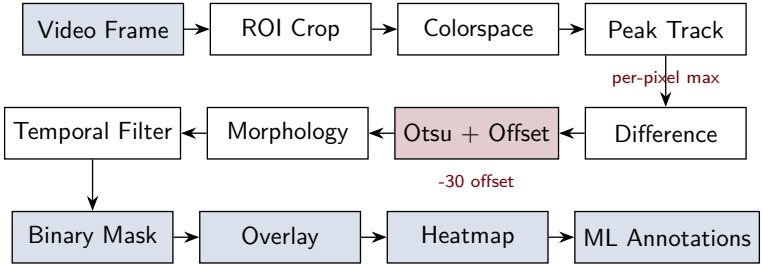
- Resin infuses dry fiber reinforcement under vacuum
- Flow front progression must be tracked
- Manual monitoring is labor-intensive
- Lighting variations cause false detections



Key Insight:

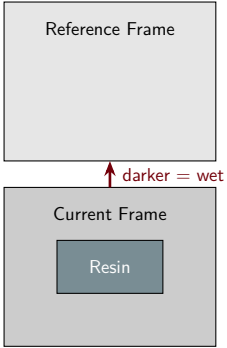
- Dry fabric $\rightarrow$ **lighter** (reflects light)
- Wet fabric $\rightarrow$ **darker** (absorbs light)

VRIFA Processing Pipeline



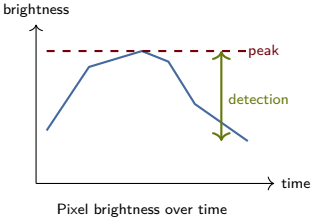
Key Innovation: Darken-Only + Peak Reference

Darken-Only Detection



Only detects **darkening** pixels
Ignores: specular highlights, auto-exposure

Peak Brightness Reference



Each pixel compared to its **historical maximum**
Handles: shadows, lighting stabilization

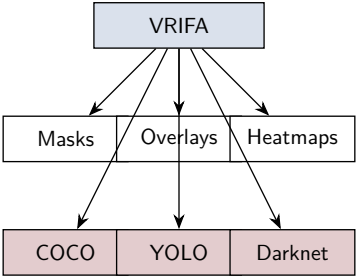
Multi-Format Outputs

Visualization Outputs

- **Binary Masks** – raw detection
- **Overlays** – red edge on original
- **Heatmaps** – temporal accumulation
- **Videos** – MP4 compilations

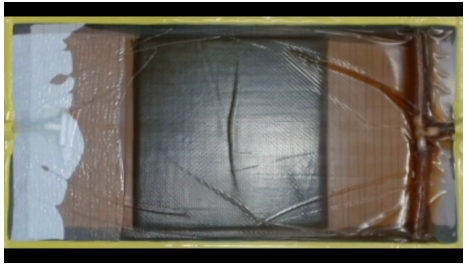
ML Training Datasets

- COCO – Detectron2, MMDetection
- YOLOv5/v8 – Ultralytics
- Darknet – YOLOv3/v4



Demo: Processing Results

Input Frame



Frame 48 – Early infusion

Later Frame

Frame 88 – Advanced flow front

```
python vrifa.py --video-path input.mp4 --write-videos
```

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Features

Darken-only detection | Peak brightness tracking
Adaptive thresholding | Multi-format ML export
Temporal filtering | Configurable ROI

`github.com/j-vaught/vrifa`